Lecture Scribe CSE400

L15_L17.pdf
PDF

The output must be a lecture scribe. It is exam-oriented reference material. It must be faithful to the lecture. The model must not invent content.

It should include definitions and notation, assumptions and conditions, statements of theorems or results, proofs or derivations step-by-step, worked examples with intermediate steps, and logical flow matching the lecture scribe.

Restrictions are to emphasizes discipline. It should be stated that there must be no intuition or commentary not in the lecture, no new examples, no simplifications or tutorials, and no invented content. Everything must be traceable to context.

The way to organize the scribe is it should specify the use of clear academic sections, following the lecture order, and following the structure of the example scribe if provided.

The scribe must be based only on the provided lecture material (slides, transcript, or notes). If something is unclear or missing in the lecture, it should not be filled in or guessed.

Symbols, equations, and notation must be preserved exactly as used in the lecture. Derivations must be shown line by line where they appear in the lecture.

Headings and subheadings must be used clearly. Statements of theorems, definitions, and examples must be kept clearly separated.

All intermediate steps shown in the lecture must be included. Steps that were explicitly written or explained must not be compressed.

The typical structure is:

Title

Introduction / Scope

Definitions and Notation

Results / Theorems

Proofs / Derivations

Worked Examples

Below is a **lecture scribe strictly based on the provided lecture slides**.
No additional explanation, intuition, or invented material has been added.
All notation, equations, and steps follow the lecture order.
All content is traceable to the provided lecture material.

CSE400 – Fundamentals of Probability in Computing

Lecture L15–L18: Joint Random Variables and Joint Distributions

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1. Introduction / Scope

In science and in real life we are often interested in **two (or more) random variables at the same time**.

Examples given in the lecture include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease

- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

Additional application examples discussed in the lecture:

Example 1: Smart Transportation

Two random variables are defined:

- X = Vehicle Speed
- Y = Traffic Density

Question posed in the lecture:

> Can speed be high when traffic density is also high?

Applications mentioned:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Throughput depends on SNR, but may also be affected by network congestion.

Applications mentioned:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

- Weather prediction
  - X = Rainfall
  - Y = Temperature
- Drone delivery
  - X = Wind speed
  - Y = Delivery time

- Rolling two dice where conditions such as $X + Y = 7$ or $X > Y$ may be studied.

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2. Discrete Case – Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X = \{x_1, x_2, \dots, x_n\}$$

$$Y = \{y_1, y_2, \dots, y_m\}$$

The ordered pair (X, Y) takes values in the product set

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

Definition (Joint PMF)

The joint probability mass function is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This gives the probability of the **joint outcome occurring simultaneously**.

Joint PMF Table Representation

The joint PMF can be organized in a table.

X\Y	y_1	y_2	...	y_j	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_j)$	...	$p(x_1, y_m$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_j)$	...	$p(x_2, y_m$
...	...	...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_j)$	...	$p(x_n, y_m$

Key Properties

$$0 \leqslant p(x_i, y_j) \leqslant 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

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3. Example – Rolling Two Dice

Experiment

Roll two fair dice.

Define:

- X = value on the first die
- T = total (sum) of both dice

Each outcome (i, j) has probability

$$\frac{1}{36}$$

A joint probability table of (X, T) is constructed.

Observation

- Table entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**.

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4. Continuous Case – Joint PDF

The continuous case is analogous to the discrete case.

- Discrete values → continuous intervals
- Joint PMF → Joint PDF
- Sums → Double integrals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) takes values in

$$[a, b] \times [c, d]$$

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\,dy) \approx f(x, y)\,dx\,dy$$

Key Properties

$$f(x, y) \geqslant 0$$

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

Note from the lecture:

> The joint PDF $f(x, y)$ may take values greater than 1.
> It is a probability density, not a probability.

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5. Worked Example

Given

$$f(x, y) = 4xy, \quad 0 \leqslant x \leqslant 1, \quad 0 \leqslant y \leqslant 1$$

Tasks:

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$.

(X, Y) lies in the unit square

$$[0, 1]^2$$

Step 1 – Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 4y \int_0^1 x \, dx$$

$$= 4y \left[\frac{x^2}{2}\right]_0^1$$

$$= 4y \left(\frac{1}{2}\right)$$

$$= 2y$$

Now integrate with respect to y:

$$\int_0^1 2y \, dy$$

$$= [y^2]_0^1$$

$$= 1$$

Hence

$$f(x, y) \geqslant 0$$

and total probability is 1.

Therefore $f(x, y)$ is a **valid joint PDF**.

Step 2 – Compute $P(A)$

Event:

$$A = \{X < 0.5, \ Y > 0.5\}$$

$$P(A) = \int_{0}^{0.5} \int_{0.5}^{1} 4xy\, dy\, dx$$

First integrate with respect to y:

$$\int_{0.5}^{1} 4xy\, dy$$

$$= 4x\,[\frac{y^2}{2}]_{0.5}^{1}$$

$$= 2x(1 - 0.25)$$

$$= 2x(0.75)$$

$$= \frac{3}{2}x$$

Now integrate with respect to x:

$$\int_{0}^{0.5} \frac{3}{2}x\, dx$$

$$= \frac{3}{2}\,[\frac{x^2}{2}]_{0}^{0.5}$$

$$= \frac{3}{2}\,(\frac{0.25}{2})$$

$$= \frac{3}{16}$$

Thus

$$P(A) = \frac{3}{16}$$

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6. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation used:

$$X \leqslant x,\ Y \leqslant y$$

which denotes the event

$$\{X \leqslant x \text{ and } Y \leqslant y\}$$

Definition

The joint cumulative distribution function is

$$F(x, y) = P(X \leqslant x, Y \leqslant y)$$

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7. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$,

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering the Joint PDF

Using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x\, \partial y} F(x, y)$$

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8. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$,

$$F(x, y) = \sum_{x_i \leqslant x} \sum_{y_j \leqslant y} p(x_i, y_j)$$

Interpretation

Add all probabilities:

- rows with $x_i \leqslant x$
- columns with $y_j \leqslant y$

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9. Properties of the Joint CDF

The joint CDF $F(x, y)$ must satisfy the following properties.

Property 1

$$F(x, y) \text{ is non-decreasing}$$

If x or y increases, then $F(x, y)$ must stay constant or increase.

Property 2

$$F(x, y) = 0$$

at the **lower-left corner of the joint range**.

If the lower-left corner is $(-\infty, -\infty)$,

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

Property 3

$$F(x, y) = 1$$

at the **upper-right corner of the joint range**.

If the upper-right corner is (∞, ∞),

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

If you want, I can also convert this into a **perfect exam-ready 8–10 page PDF lecture scribe (exact format professors expect in probability courses)** so it's easier to study for your mid-sem.